

Omega-N: Interpretable Structural Node Descriptors and Their Applicability Domain

Alberto Acedo
Biome Makers Inc

Abstract

A composite structural index summarises a whole network in one number. For a triangle-based index that number is spectrally redundant: $\text{Tr}(A^3)$ is exactly the third moment of the adjacency spectrum, so it cannot carry information that a spectral statistic does not already have. The non-redundant content sits one level down, in the node-wise decomposition $\text{diag}(A^3)$, which depends on eigenvectors and is not determined by the spectrum: cospectral graphs with different per-node triangle counts exist [1]. That distinction is not new here. It was stated as Corollary 1 of the theory paper for this index family, together with an explicit prediction: the global scalar should tie sharpened spectral baselines rather than beat them, while the node-wise attribution should do better where the number of structural epicentres is unknown [1]. This paper is the test of that prediction.

We construct **Omega-N**, a node-level descriptor obtained by localizing each of the four factors of the index, and we show that the direct localization is badly conditioned: two factors degenerate on the most ordinary nodes rather than on extreme ones. Two corrections fix it, both taken from published practice: every local factor is expressed as an excess over a degree-preserving null, and the neighbourhood is replaced by a personalized-PageRank distribution at several scales [2]. The result is ten interpretable features per node, computed from the graph alone, with no attributes, no training and no embeddings.

The prediction holds in the direction it was stated. Against a recursive feature engine taken to five levels of recursion, Omega-N wins on one and ties on four of six in-domain node-classification evaluations, using ten features against up to 252, at one to two orders of magnitude less cost than a diffusion-wavelet baseline, with a declared null returning chance for every arm. Domain membership is decided by two statistics computed from the graph and the labels but not from the descriptor’s performance; that rule partitions the eight benchmark networks without error, and the two it excludes are the two on which the descriptor loses.

The strongest application is drug-target prioritisation over protein interaction networks: +0.073 to +0.144 AUPRC over a centrality battery across four network constructions, replicated on an independent affinity-purification network and with an independent label source, surviving three bias controls including one that rebuilds the graph without its text-mining channel (degree-matched, ten repetitions: +0.1047 and +0.1030, both 10/10, $p = 0.00195$). The clearest negative sits in the same application: adding Omega-N to a published pipeline of centralities plus Node2Vec changes nothing (+0.0014 AUPRC, $p = 0.31$), and on an unbiased experimental network the entire centrality battery sits at chance, which places most of the absolute signal in curation rather than in biology. The claim we defend is narrow and explicit: ten interpretable features without training improve on hand-crafted centralities, not on learned representations.

1. Introduction

Composite structural indices are read as scalars, and the scalar is the weakest object in the framework that produces it. For the index family this paper works with, that weakness has a precise form. The global functional is exactly the third moment of the adjacency spectrum, which places it in the same family as the absorption ratio, effective rank and spectral entropies; any two functionals of the same eigenvalues cannot be expected to carry mutually exclusive information [1]. The node-wise decomposition is different in kind. Writing A as a sum over eigenpairs, the per-node quantity depends on the eigenvectors, and cospectral graphs with different per-node triangle counts exist, so no eigenvalue-based scalar can reproduce it [1].

The theory paper states the consequence as its Corollary 1 and then, unusually, as a falsifiable prediction: the global scalar should tie sharpened spectral baselines rather than beat them, while the node-wise attribution, being parameter-free where eigenvector attribution requires committing to a number of components, should do better when the number of structural epicentres is unknown [1]. It also says that the prediction had not been tested. Testing it requires three things that the corollary does not supply: a localization that is numerically well behaved, a demonstration that the resulting coordinates are not re-statements of quantities already available, and a statement of the conditions under which the localization helps. Those three requirements organise this paper, and the result is what the corollary anticipated: a narrow win, four ties, and two losses that fall on the wrong side of a stated condition.

Vectors of structural quantities per node are not new. Guimera and Amaral’s within-module degree and participation coefficient date from 2005 [3]; RolX and its ReFeX feature engine build node roles by recursive neighbourhood aggregation [4]; struc2vec [5] and GraphWave [6] learn structural embeddings; and there is an established evaluation framework for the family [7]. Local versions of global network statistics have their own line, from naive local assortativity [8] to the multiscale personalized-PageRank localization of Peel, Delvenne and Lambiotte, which also observed that naive local estimates overfit at low degree [2]. What is new here is not the idea of a vector per node. It is the localization of one specific composite whose sign structure is inherited from an empirical gradient, together with the null calibration that makes its four local factors commensurable, and the applicability rule that says when the exercise is worth doing.

Three lines of recent work are close enough that the distinction should be explicit. Homomorphism-count node embeddings report results that do not match state-of-the-art neural architectures while remaining comparable to other graph learning models, and locate their contribution in explainability at the level of the individual feature [9]. Our claim has the same shape, and we state it as plainly. Invariant-based diagnostics for graph benchmarks show across 26 datasets that permutation-invariant, task-agnostic descriptors characterise structural heterogeneity, and that simple invariant-based models are competitive with transformer and message-passing baselines, concluding that expressivity is not the main driver of predictive performance [10]; that work asks what benchmarks measure, where we ask what one composite index measures once localized. And an unsupervised evaluation framework for this family [41] judges structural embeddings by whether they reconstruct six standard node features, a set essentially identical to the centrality battery we use as our rival, which makes our comparison one against the field’s own reference set rather than one of our choosing.

One objection deserves an answer here rather than in a rebuttal. Fast structural-embedding methods criticise approaches that require computing several complicated graph metrics, such as PageRank, structural holes or the clustering coefficient, as ad hoc and expensive [11]. The criticism lands on any hand-built descriptor, including this one. The answer is not that the objection is wrong but that it describes a trade: ten interpretable coordinates carrying no training against a learned representation that performs better and explains less. We measure where that trade stops paying, and it stops precisely against learned representations.

A second contribution emerged from the same property and is worth stating in the introduction because it is not an application of the descriptor so much as a use for its defining feature. Because the spectral gap is a functional of the spectrum while per-node triangle counts are not, two graphs can be built to agree on the gap and disagree on local triadic organisation. Whole-brain modelling needs exactly that construction: a recent proposal identifies the spectral gap as the operative structural quantity distinguishing two dynamical regimes, and states in its own limitations that a null model matched for the gap is required before the effect can be attributed to organisation rather than to the size of the gap [12]. We supply the construction and verify that it separates, without claiming anything about the brain.

2. The descriptor

2.1 Definition of the descriptor

We work with the index of [1],

$$\Omega = \frac{C \cdot D}{M \cdot \text{Coex}},$$

where C is the global clustering coefficient, D the connection density, $M = 1/\lambda_2(L)$ the inverse algebraic connectivity with $L = \text{diag}(\mathbf{k}) - A$ the graph Laplacian, and $\text{Coex} = \text{Var}(\mathbf{k})$ the degree variance.

Definition 1 (Omega-N). Let A be the adjacency matrix of an undirected graph on n nodes with m edges, let $\mathbf{k} = A\mathbf{1}$ be the degree vector, and write $s^{(1)} = A\mathbf{k}$ and $s^{(2)} = A\mathbf{k}^{\odot 2}$ for the first two neighbour-degree moments. Omega-N assigns to each node i four base coordinates.

Degree share, the node’s contribution to the density:

$$d_i = \frac{k_i}{n-1}, \quad \frac{1}{n} \sum_i d_i = D.$$

Triadic excess, with t_i the triangles incident on node i and E_i its expectation under the configuration model [21]:

$$E_i = \frac{(s_i^{(1)})^2 - s_i^{(2)}}{4m}, \quad c_i = \frac{t_i - E_i}{E_i + 1}.$$

The unit offset keeps the ratio finite where the null expectation approaches zero. Expressing a local count as a deviation from a degree-preserving null is standard practice in motif analysis, where the abundance score takes the form (observed - expected)/(observed + expected + epsilon) [27]; the coordinate above is that device with a different normalisation, adopted rather than proposed.

Neighbourhood dispersion, referred to its own null expectation:

$$h_i = \log \left(1 + \frac{\text{Var}_{j \sim i}(k_j)}{\langle k^2 \rangle / \langle k \rangle} \right).$$

Spectral energy, the node’s term in the Dirichlet energy of the Fiedler vector $\mathbf{v}$ of the Laplacian [22]. Note that these terms sum to λ_2 , that is to $1/M$ and not to M ; the local factor corresponding to the third global factor is therefore the inverse of that factor, which fixes the sign it carries in Proposition 2:

$$e_i = \frac{1}{2} \sum_j A_{ij} (v_i - v_j)^2, \quad \sum_i e_i = \lambda_2(L).$$

The coordinates c , h and e are then smoothed by the personalized-PageRank operator P_α at $\alpha \in \{0.5, 0.9\}$ following [2], giving $4 + 3 \times 2 = 10$ features per node.

Remark 1 (three non-degree coordinates, not four). The first coordinate *is* the degree. We keep it because the composite it comes from contains it, and because excluding it would make the comparison against degree-based rivals harder to read, but the descriptor contributes three non-degree quantities, not four, and the comparisons in Section 4 are read accordingly.

Remark 2 (why the fourth factor is replaced). The literal fourth factor of the scalar index is the degree variance, whose per-node contribution $(k_i - \bar{k})^2$ is a function of the degree alone and vanishes at the mean degree. Proposition 1 shows what that does to the localization; coordinate 3 replaces it with a quantity that measures the same thing, heterogeneity in the neighbourhood, without the degeneracy.

2.2 Conditioning failures of the direct localization

Proposition 1 (Conditioning failure of the naive localization). Localizing the scalar index by substituting per-node shares into the multiplicative composition produces two degeneracies, and both occur on ordinary nodes rather than extreme ones. The degree-variance share vanishes at the mean degree, so the contraction diverges there; and the clustering share is zero on every triangle-free node, so a log-contraction is minus infinity there.

Measured. On seven graphs (Karate, Florentine, Davis, Les Miserables, and Erdos-Renyi, Barabasi-Albert and Watts-Strogatz controls): in a Watts-Strogatz graph 36 of 62 nodes degenerate and the ratio $\text{mean}(\Omega_i)/\Omega$ reaches 2.9×10^{11} ; in Les Miserables 17 of 77 nodes are triangle-free. The practical consequence, which we accept rather than work around, is that the publishable object is the vector, not its contraction back to a scalar.

Proposition 2 (Exact aggregation identity). Define each global factor as the arithmetic mean of its node shares, and write $J_X = \log \langle X \rangle - \langle \log X \rangle \geq 0$ for the log arithmetic-to-geometric gap of factor X . Then

$$\log \Omega - \log \text{GM}(\Omega_i) = J_C + J_D + J_{\lambda_2} - J_{\text{Coex}}$$

exactly, where GM is the geometric mean of the node-level indices and J_{λ_2} is the gap of the third factor in its localized form, $\lambda_2 = 1/M$, as defined above.

Verified numerically to $4e-16$ on six graphs. Three consequences follow, and the first two correct claims made in an earlier draft of this work.

- There is no general inequality between the scalar and the aggregate of its parts: three non-negative terms minus one, so the sign is indeterminate.
- Equality does not hold on regular graphs. Only J_D and J_{Coex} vanish there; local clustering still varies node to node, and in a random 6-regular graph $J_C = 9.66$.
- The right aggregate is the geometric mean of the node indices, not the arithmetic one, which also explains a divergence observed in the financial application of Section 5.3.

Remark 3 (the identity motivates the reformulation independently). J_C is dominated by the floor imposed on triangle-free nodes rather than by genuine heterogeneity, reaching 13.5 on the Florentine graph purely because some nodes have zero clustering. Under the naive localization, one of the four terms of an otherwise exact identity is therefore meaningless. Note also that Proposition 2 is a statement about the original multiplicative composition; the descriptor of Definition 1 abandons that composition and is a feature block, not a product, so the identity does not apply to it.

2.3 Non-redundancy with the degree

Rank and linear correlation of the naive contraction with degree across the seven graphs: 0.20 to 0.65 (Pearson), 0.48 to 0.81 (Spearman). The reference point is that the raw per-node triangle count correlates 0.996 with degree in the financial application of the same framework [13]. The composition separates where the raw numerator did not, and the mechanism is identifiable: local clustering is anticorrelated with degree, at -0.53 on Karate.

2.4 Independence from discrete Ricci curvature

The framework relates triangle counts to Ollivier-Ricci curvature through an upper bound [1], so a reader who knows it will ask whether the descriptor is curvature under another name. It is not, and the answer is measurable rather than argued.

Ollivier-Ricci curvature [23] with uniform neighbour measures and shortest-path transport cost, the transport problem solved exactly by linear programming per edge, averaged over the edges incident on each node. Rank correlation against the triadic-excess coordinate, and against the degree for reference:

network	n	mean degree	$\rho(\text{curv.}, \text{triadic excess})$	$\rho(\text{curv.}, \text{degree})$
brazil-airports	131	15.3	-0.118	+0.787
europa-airports	399	30.0	-0.181	+0.659
usa-airports	1,190	22.9	+0.034	+0.485
minesweeper	10,000	7.9	-0.023	-0.388
tolokers	11,758	88.3	+0.182	-0.195
amazon-ratings	24,492	7.6	+0.096	-0.165
questions	48,921	6.3	+0.099	-0.253
roman-empire	22,662	2.9	+0.460	-0.268

On seven of the eight networks the triadic-excess coordinate is essentially uncorrelated with curvature, $|\rho|$ at most 0.18, and the sign is not stable: negative on the three airport graphs and on minesweeper, positive on the others. The degree behaves differently again, correlating strongly and also changing sign. The two are independent coordinates rather than two names for one quantity.

The exception is informative. On roman-empire the correlation reaches +0.460, and roman-empire is the sparsest graph in the set at mean degree 2.9. Where there is almost no triadic structure the triadic excess has little of its own to say and tracks whatever local geometry remains. Independence from curvature is a property of the regime in which the descriptor is useful, not one it holds everywhere, and Section 4.2 makes that regime explicit.

Approximations, declared. On the four large graphs we sample 800 edges, cap each neighbourhood at 40 sampled neighbours, and evaluate inter-neighbour distances as 0, 1, 2 or 3 by direct adjacency lookup, capping at 3; the transport problem itself remains exact. The distance cap can only underestimate long distances, in the same direction for every node, and these graphs have small diameter. The four small graphs use exact shortest paths and no sampling.

2.5 Predictive validity of the non-degree coordinates

A structural coordinate that is independent of the degree may still be independent of anything that matters. We test against a node attribute that is not structural, with identical folds for every feature set and with two mandatory rivals rather than degree alone: the standard centrality battery, and the Guimera-Amaral pair [3].

Datasets are the Twitch social graphs (PT-BR, ES, EN-GB) with audience size as target. The Wikipedia chameleon and squirrel graphs are excluded and the exclusion is declared: Platonov et al. report duplicate-node leakage in them [14], and we verified that 45.1% of chameleon nodes and 36.2% of squirrel nodes share an identical neighbourhood and an identical target with another node.

Out-of-sample R^2 , with the ten-feature descriptor of Definition 1 at the released 20-term setting. Five-fold cross-validation, the same folds for every arm on a given graph, with the same random forest as elsewhere; the parenthesised figure is the standard error across the five folds.

graph	n	degree	battery	Guimera-Amaral	Omega-N	Omega-N + battery
Twitch	1,912	0.468 (.021)	0.517 (.018)	0.435 (.018)	0.583 (.015)	0.603 (.017)
PT-BR						
Twitch ES	4,648	0.419 (.016)	0.448 (.013)	0.380 (.014)	0.539 (.016)	0.557 (.015)
Twitch	7,126	0.327 (.018)	0.361 (.024)	0.299 (.021)	0.410 (.017)	0.431 (.021)
EN-GB						

The margins over the battery are 0.066, 0.091 and 0.049. Two of the three, PT-BR and ES, sit between roughly four and seven fold standard errors; the third, EN-GB, is around two, comparable to the node-classification win of Section 4.2. The dispersion quoted is across folds and is therefore optimistic, for the reason given in Section 3.3; it is used here only as a scale, not as a test. Omega-N beats degree, the battery and Guimera-Amaral on all three graphs, and the union with the battery is better still, so the descriptor is complementary to it rather than redundant with it.

3. Experimental design

The descriptor is evaluated in three fields chosen because they differ in how their graphs are built, not only in size: node classification on published benchmarks, systemic-risk attribution on financial correlation graphs, and drug-target prioritisation on protein interaction networks. Each subsection states its graphs, its label, its rivals and its protocol, because those differ by field and stating them once in the aggregate would hide the differences that matter.

Two conventions are common to all three. Every network is restricted to its largest connected component; without that restriction λ_2 is zero and the spectral coordinate of Definition 1 degenerates into a component indicator, a failure that is invisible in the output. And every arm within a field uses the same classifier and the same folds, so that differences are attributable to the features.

3.1 Node classification

The graphs are the five heterophilous benchmarks of Platonov et al. with their ten official splits [14], and the three air-traffic graphs of Ribeiro et al. with labels given by activity quartile [5]. *minesweeper* serves as a declared null: its labels are independent of structure by construction, so every arm should return chance.

The rivals are the centrality battery (degree, k-core number, PageRank and local clustering coefficient), ReFeX [4] at two, three, four and five levels of recursion (28, 60, 124 and 252 features); and GraphWave [6]. Reporting the full ReFeX depth sweep matters: at two levels our margins are larger, and the saturation at four and five levels costs us one result.

dataset	nodes	mean degree	no triangle	label
tolokers	11,758	88.3	5.0%	worker banned (binary)
europa-airports	399	30.0	6.0%	activity quartile (4)
usa-airports	1,190	22.9	20.4%	activity quartile (4)
amazon-ratings	24,492	7.6	0.8%	product rating (5)
questions	48,921	6.3	83.1%	user still active (binary)
brazil-airports	131	15.3	12.2%	activity quartile (4)
roman-empire	22,662	2.9	28.6%	syntactic role (18)
minesweeper (null)	10,000	7.9	0.0%	mine present (by design)

Our GraphWave is a reimplementation and every approximation is declared: Chebyshev polynomial of degree 30, verified to 6e-15 against a dense matrix exponential on Karate; two scales at the endpoints of the published heuristic; a 25-point characteristic-function grid on [0,100]; the empirical coefficient distribution estimated on a fixed 2,048-row subsample; and, on the two largest graphs, a declared 4,096-node evaluation subsample with the same nodes for every arm. We report it under both the arbitrary fixed scales we first used and the published heuristic, and take the better of the two as the rival.

The classifier is a random forest [29] throughout (300 trees, minimum leaf size 5, balanced subsample weighting for imbalanced classes). Official splits where they exist; twice-repeated stratified five-fold cross-validation on the air-traffic graphs. AUROC for binary targets, accuracy for multi-class.

3.2 Systemic-risk attribution in financial correlation graphs

Two universes are used: 42 diversified assets over 2006-01 to 2026-07, giving 5,166 trading days, and 447 S&P 500 constituents with complete history, 2012-08 to 2017-08, giving 1,258 trading days. $A = |\text{corr}|$ with zero diagonal, estimated in-window; filtered variants are the Mantegna minimum spanning tree [25] and the planar maximally filtered graph [26]. These graphs are complete by construction and the filtered variants connected by definition.

The target is forward marginal expected shortfall: an asset’s mean loss on the 5% worst days of the equally weighted system, over the next non-overlapping window.

The weighted variant is defined as follows. With $\mathbf{s}$ the strength vector and $S = \sum_i s_i$, the strength-based configuration model $p_{ij} = s_i s_j / S$ gives the closed form

$$E_i = \frac{s_i^2 \left(\sum_j s_j^2 \right)^2}{S^3}, \quad \text{score}_i = \frac{(A^3)_{ii} - E_i}{E_i + \varepsilon},$$

defined only where $(A^3)_{ii} > 0$.

The rivals are the asset’s own past MES, its beta to the system, and its strength in the graph. Partial rank correlation with forward MES controlling past MES and beta; 19 non-overlapping windows with a sign test, plus a block bootstrap over 234 overlapping windows. Controls and predictor are estimated on the same 250-day window, which matters: a control window shorter than the estimation window measures history rather than structure.

3.3 Drug-target prioritisation on protein interaction networks

Four network constructions of increasing purity are used, three of them thresholded from STRING [31] at successively narrower channel sets and one from BioPlex [32], so that the contribution of curation can be separated from that of experiment.

A fifth candidate, HuRI [33], was screened and excluded before the comparison: as a systematic two-hybrid map it records binary interactions rather than complexes, and 58.7% of its nodes have no triangle at all, so there is no triadic substrate to measure. The criterion was fixed in advance and the figure is reproducible; Section 6 returns to what that exclusion costs in credibility.

The label is approved-drug-target status, taken from Open Targets [20] (13,307 target-drug pairs, 1,058 targets at APPROVAL stage), mapped to STRING identifiers through the alias table and to BioPlex through gene symbols. This label is independent of the one used in an earlier version of this analysis

network	construction	nodes	mean degree	positives
STRING, combined	all channels incl. text mining, score ≥ 700	15,882	29.8	6.6%
STRING, no text mining	experimental + database channels	10,987	20.7	8.9%
STRING, experimental only	experimental channel	7,414	17.7	9.3%
BioPlex 293T	AP-MS, independent of STRING	13,923	17.0	5.5%

(DGIdb [36]): the two agree on 1,002 genes but DGIdb lists 3,424 and Open Targets 1,052, a Jaccard index of 0.288. The network is replicated on BioPlex and the label is replicated here, so neither is a single point of failure.

The rivals are the centrality battery, extended to six with eigenvector centrality and neighbour-degree sum; and the published pipeline of six centralities plus 64-dimensional Node2Vec embeddings [24] (10 walks of length 40, skip-gram, window 5, 3 epochs) fed to gradient-boosted trees (400 estimators, depth 6, learning rate 0.08, subsample 0.8, positive-class weighting by the class ratio).

AUPRC is read as primary [30] because positives are 5.5% to 9.3% of nodes and AUROC is optimistic at those rates. Significance is assessed over ten independent repetitions, each with its own split seed and its own degree matching, with a paired Wilcoxon test across repetitions rather than across folds; folds within a repetition are not independent and testing across them would inflate the result. With ten paired observations the attainable floor is $p = 0.00195$.

Degree matching addresses a specific concern. Where the concern is that a margin merely restates the degree, each positive is matched to a negative of nearly identical degree and the comparison runs on the matched subset only, where the degree itself falls to chance. Matching is redrawn in every repetition so that its randomness enters the error.

3.4 Implementation

All figures were produced with the settings of the released implementation, including the 20-term truncation of the personalized-PageRank smoothing. An earlier draft used 25 terms, worth 0.785 rather than 0.776 AUROC on tolokors and turning one tie into a win; we regenerated the tables at the released setting rather than keeping the more favourable one. The spectral coordinate uses LOBPCG, which does not always reach the requested tolerance at these sizes; the released implementation warns when the input has more than one component.

4. Results

We report the three fields in order of outcome: the application where the descriptor helps most, the benchmark where it ties, and the field where it closes. Within each, the result is stated before its qualifications.

4.1 Drug-target prioritisation

Omega-N against the centrality battery, AUPRC:

network	n	mean degree	base rate	battery	Omega-N	margin
STRING, combined	15,882	29.8	6.6%	0.2891	0.4007	+0.112
STRING, no text mining	10,987	20.7	8.9%	0.3870	0.5306	+0.144
STRING, experimental only	7,414	17.7	9.3%	0.4147	0.4966	+0.082
BioPlex 293T	13,923	17.0	5.5%	0.0934	0.1668	+0.073

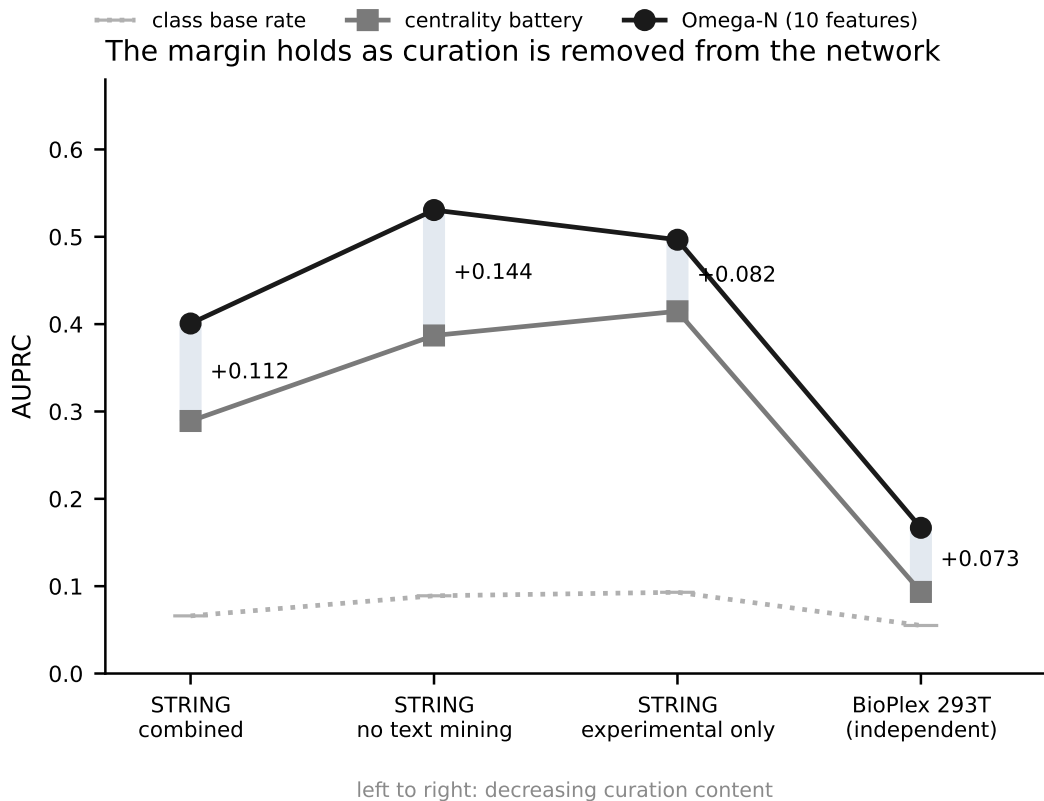


Figure 1: The margin survives the removal of curation; the absolute performance does not. AUPRC for approved-drug-target prediction across four protein interaction networks, ordered by decreasing curation content. The gap between battery and descriptor stays between +0.073 and +0.144 while the battery itself falls from 0.41 to 0.09. On BioPlex the battery is close to the class base rate.

Ten independent repetitions with per-repetition degree matching: on STRING without text mining the difference is +0.1047 (95% CI [+0.0968, +0.1127], 10/10, Wilcoxon $p = 0.00195$); on BioPlex +0.1030 ([+0.0955, +0.1106], 10/10, $p = 0.00195$). The two margins are almost identical although the networks share no construction, no experimental technique and no curation.

The three analyses that follow were run under the earlier label (DGIdb) and are reported at that label: the bias controls, the cut-off sweep and the quantification of the connectivity defect. They are not directly comparable with the margins in the table above, which use Open Targets. We report them because their conclusions concern the robustness of the margin rather than its size, and because regenerating them under the new label would not change what they establish.

Three bias controls were applied and the margin survives all three. Study intensity, proxied by the length of the curated functional annotation, is real (targets 394 characters against 302, Spearman +0.375 with

degree, AUPRC 0.258 on its own) and was given to the rival: the battery rises from 0.4677 to 0.4820 and Omega-N still wins. Degree-matched case-control, where degree collapses to AUROC 0.424: the margin holds. And the network rebuilt without the text-mining channel, then without curation as well: the margin holds and widens slightly.

The result does not depend on the confidence cut-off. That cut-off was fixed at 700 throughout and swept afterwards, again under the earlier label. Across 400, 550, 700, 850 and 900, with mean degree falling from 95.4 to 16.4, the margin is +0.085, +0.081, +0.071, +0.066 and +0.069 AUPRC. It never changes sign and never depends on the cut-off.

One defect surfaced while running the sweep, and it is worth reporting. Forcing the eigensolver to converge revealed that λ_2 collapses to order 10^{-11} on the graph as first built: the network is disconnected (126 components at cut-off 700), so the Fiedler vector was a component indicator and the fourth coordinate contributed nothing. Restricted to the largest connected component (15,882 of 16,201 nodes) λ_2 is 0.0276 and the coordinate is meaningful again. The corrected margin at cut-off 700 is +0.0698 against +0.0707 with the defect present. The effect is negligible and in the direction that matters: every number above was obtained with one of the four coordinates effectively inert, so the tables understate the descriptor. We report the defect because a reader reproducing the code would otherwise find the discrepancy.

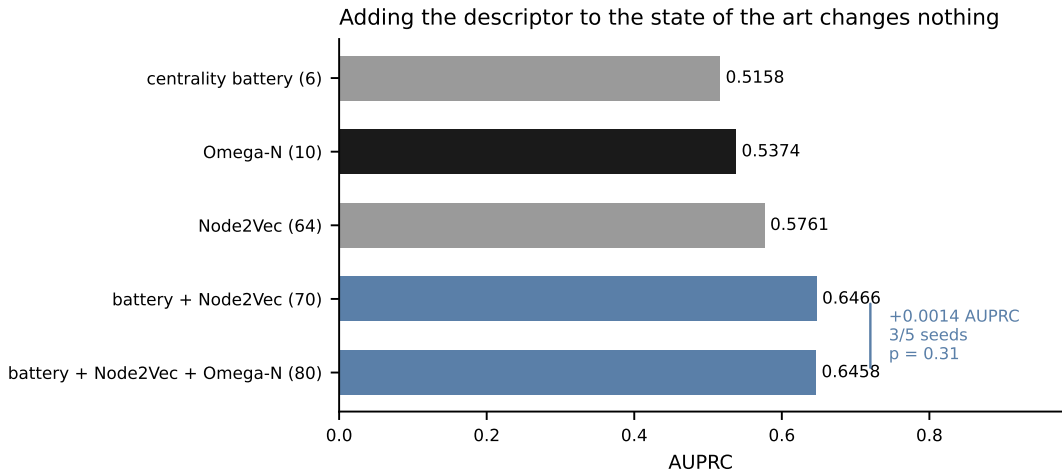


Figure 2: The descriptor does not improve on learned representations. AUPRC on curated STRING under a single gradient-boosting classifier and identical folds. Bars are a single seed, on which the eighty-feature arm scores marginally below the seventy-feature one; the quoted +0.0014 is the mean over five seeds, and the difference is positive on three of the five. Node2Vec alone already outscores Omega-N.

The negative result sits in the same application. Against the published pipeline on curated STRING:

arm	features	AUROC	AUPRC
centrality battery	6	0.7833	0.5158
Node2Vec	64	0.8235	0.5761
battery + Node2Vec	70	0.8463	0.6466
Omega-N	10	0.7936	0.5374
battery + Node2Vec + Omega-N	80	0.8458	0.6458

The table reports one seed, on which the eighty-feature arm scores 0.6458 against 0.6466; over five seeds the mean difference is +0.0014 AUPRC, positive on three seeds and negative on two, Wilcoxon $p = 0.31$. Either way, adding Omega-N to the state of the art changes nothing. Node2Vec alone already outscores Omega-N, so whatever the ten coordinates add over hand-built features is already inside the embedding.

Two further findings cut against any practical reading. On BioPlex, an unbiased AP-MS network, degree alone scores AUROC 0.5007 and the whole centrality battery 0.5099: raw experimental topology does not predict drug-target status. Omega-N reaches 0.5602, better than chance and not a prioritisation tool. The 0.79 obtained on curated STRING therefore reflects accumulated knowledge embedded in the network rather than structure discovered de novo.

4.2 Node classification and the applicability domain

Air-traffic graphs, twice-repeated stratified five-fold, same folds for every arm:

	brazil (131)	europa (399)	usa (1,190)
degree alone	0.748	0.564	0.552
ReFeX, 2 levels (28)	0.771	0.569	0.682
ReFeX, 3 levels (60)	0.771	0.589	0.685
ReFeX, 4 levels (124)	0.786	0.586	0.681
ReFeX, 5 levels (252)	0.786	0.576	0.678
GraphWave, fixed scales (100)	0.781	0.571	0.628
GraphWave, published heuristic (100)	0.695	0.551	0.613
Omega-N (10)	0.656	0.592	0.680

Platonov graphs, official splits:

feature set	minesweeper (null)	tolokers	amazon- ratings	questions	roman-empire
degree	0.501	0.551	0.365	0.605	0.211
standard	0.497	0.704	0.388	0.614	0.266
battery					
ReFeX, 2-5 levels	0.493	0.761-0.769	0.443	0.681	0.339
naive 1-hop localization (4)	0.497	0.693	0.381	-	-
Omega-N (10)	0.494	0.776	0.442	0.684	0.322
Omega-N + battery	0.486	0.791	0.450	-	-

roman-empire is one of the two losses and is reported here rather than only asserted: 0.322 against ReFeX's 0.339, on the graph the applicability rule excludes.

Two controls each cost us a result. Taking ReFeX to four and five levels brings it to 0.685 on usa against 0.680, turning a win into a tie; on brazil deeper ReFeX reaches 0.786, so the loss there is against a stronger rival than first measured. Separately, aligning the tables with the released code moves europa from 0.611 to 0.592 against ReFeX's 0.589, turning a second win into a tie.

GraphWave is reported under both scale settings. On discovering that our arbitrary fixed scales were 3 to 23 times smaller than the published heuristic, we recomputed; the heuristic makes GraphWave *worse* on all three airport graphs. The mis-scaling was not crippling the baseline, and our original numbers were generous to it rather than unfair. Two implementation notes must stay in: usa-airports is disconnected, so the heuristic diverges unless the smallest non-zero Laplacian eigenvalue is used, and with the naive λ_2 the embedding collapses to chance (0.249); and our λ_2 from LOBPCG disagreed with a shift-invert solve by a factor of 5 on amazon-ratings, which moves the scales by 2.3x without moving any verdict.

Runtimes were measured rather than estimated: 0.2 s against 4.7 s on usa-airports, 1 s against 145 s on minesweeper; and on a 519k-edge graph the original diffusion kernel did not finish in seven hours while the descriptor took seconds. The advantage is one to two orders of magnitude depending on the graph, not a single figure. The claim is against the *original* GraphWave: a scalable variant exists [15] and our comparison does not address it.

Two statistics decide domain membership, both computed from the graph and the labels and neither from the descriptor's performance. S1 is the mean degree: a chain-like graph gives a triadic excess almost nothing to measure. S2 is the mutual information between degree and label normalised by the label entropy: where the label is close to a function of the degree, the degree is the right answer and calibrating against a null removes signal rather than noise.

network	mean degree	$I(k;y)/H(y)$	outcome
tolokers	88.3	0.014	win (0.776 vs 0.769)
europa-airports	30.0	0.379	tie (0.592 vs 0.589)
usa-airports	22.9	0.328	tie (0.680 vs 0.685)
amazon-ratings	7.6	0.002	tie (0.442 vs 0.443)
questions	6.3	0.033	tie (0.684 vs 0.681)
minesweeper (null)	7.9	0.000	chance for every arm
brazil-airports	15.3	0.650	out of domain by S2
roman-empire	2.9	0.087	out of domain by S1

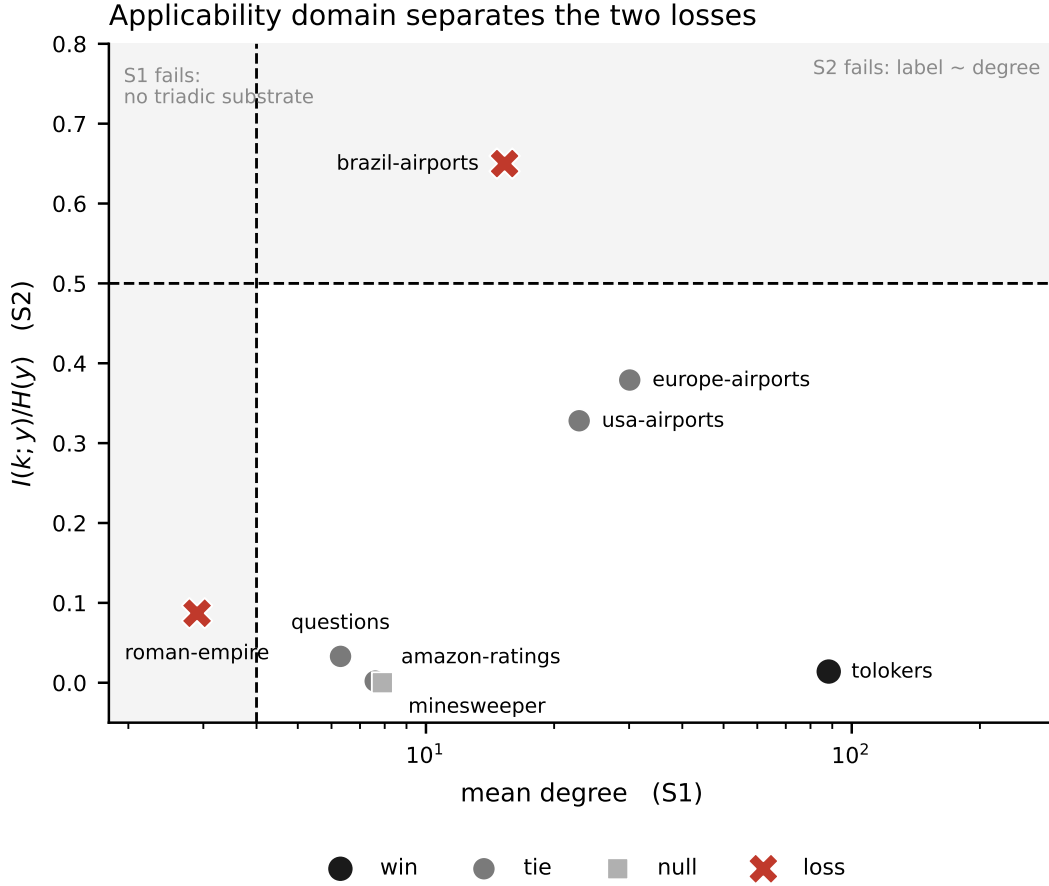


Figure 3: Applicability domain. Each point is one benchmark network, placed by the two statistics that decide domain membership: mean degree (S1, horizontal, log scale) and the mutual information between degree and label normalised by label entropy (S2, vertical). Neither is computed from the descriptor’s output. Dashed lines mark the rule; shaded regions are outside it. The rule partitions the eight networks without error.

The rule “mean degree at least 4 and $I(k;y)/H(y)$ at most 0.5” separates the eight networks without error. *brazil* fails S2 at 0.650 against 0.379 for the next highest; *roman-empire* fails S1 at 2.9 against 6.3 for the next lowest. Both gaps are close to a factor of two, and every threshold pair in $[3, 6] \times [0.45, 0.60]$ gives the same partition. S1 is the mean degree rather than the fraction of nodes without a triangle, and the distinction matters: *questions* has 83.1% of its nodes in no triangle and still falls inside the domain, because its mean degree of 6.3 leaves a minority of well-connected nodes with ample triadic structure. What excludes *roman-empire* is a low mean degree together with a chain-like topology, not the absence of triangles on its own.

Inside the domain the tally is one win and four ties, plus a clean null in which all six arms including a 100-dimensional embedding return chance. The win is narrow and we call it narrow: 0.776 against 0.769, about two standard errors. The four ties are the substantive result, because the descriptor reaches the

same place with ten interpretable coordinates and no recursion where the rival uses up to 252.

A connectivity audit closes this section. All five Platonov graphs are connected, and the correlation graphs of Section 4.3 are complete by construction, so only usa-airports differs: 1,186 nodes in the largest component and two pairs outside it. Restricted to the largest component the descriptor scores 0.7125 rather than 0.6798. We keep the conservative figure in the table and note the difference rather than quoting the more favourable number.

4.3 Systemic-risk attribution

The closure comes first. On the framework’s own synthetic epicentre generator (16 nodes, 120 simulations per regime, four regimes), the raw per-node triangle count reaches a 0.975 to 0.994 hit rate and the configuration-null score 0.884 to 0.908. It loses in all four regimes. On real data the raw count correlates +0.994 with strength, reproducing an independently published 0.996 [13], so strength is a sufficient statistic for detecting concentration and dividing by the null expectation removes exactly what must be detected. This is the third independent appearance of the S2 condition of Section 4.2.

The positive is modest and we describe it as modest. Partial rank correlation with forward MES, controlling past MES and beta:

score	correlation	independent windows	p	bootstrap 95% CI
strength (complete graph)	+0.010	10/19	1.000	[-0.037, +0.079]
raw triangle count	+0.037	-	-	-
Omega-N score	+0.142	16/19	0.0044	[+0.107, +0.216]
Omega-N score, PMFG	+0.092	15/19	0.0192	[+0.075, +0.169]

With four scores compared, a Bonferroni threshold of 0.0125 passes the complete graph and not the PMFG. Filtering the graph makes things worse, not better. Signed triangles add nothing (+0.083, 62% of windows). The second universe replicates in sign and magnitude (+0.121) while strength turns negative (-0.080), but it carries no significance: five years give four non-overlapping windows ($p = 0.625$) and its bootstrap resamples two blocks, so its interval is not calibrated and must not be read as evidence.

An artefact was found and removed, and it is reported because it is instructive. With the epsilon in the denominator and no domain restriction, a triangle-free graph returns -1 plus floating-point noise that reproduces the inverse ranking of strength; on a minimum spanning tree this gave a spurious -0.218, the exact mirror of strength. The score is now undefined on triangle-free nodes and trees are excluded by construction.

Two claims are not made here. There is no lead time: the underlying scalar coincides with stress rather than anticipating it, and nothing here changes that. No economic significance: no portfolio or capital-allocation result is reported.

4.4 Supervised graph anomaly detection

We ran the descriptor on tolokers, one of the ten datasets of the canonical supervised graph-anomaly-detection benchmark [28], where it reaches 0.776 AUROC against 0.769 for ReFeX at its best depth on the official splits, with ten features against two hundred and fifty-two. We report this as a data point and claim nothing from it. The complementarity of structural and attribute features in that field is already established with measurements [16]; the incumbent is a tree ensemble with neighbourhood aggregation that uses node attributes, which our structure-only arms do not, so the two are not comparable; and a public reproduction on Elliptic already shows that rich tabular features encode graph structure well enough that learned embeddings do not improve on gradient boosting, which is the same phenomenon we document in Section 4.1. A proper claim there would need an attribute-free graph with published splits and the incumbent run with attributes removed.

5. Discussion

5.1 Conditions underlying the positive result

The protein networks satisfy all the conditions the descriptor needs, and they are the only field tested that does. Mean degree between 17 and 30 gives ample triadic substrate. The label, whether a gene is the target of an approved drug, is not a function of connectivity: on the contrary, the literature reports that drug targets are neither hubs nor bridges, and that degree is a weak predictor of target status [34,35]. And the nodes have almost no attributes beyond taxonomy, so topology is not one source among many but nearly the only one available.

That combination explains the margin, and its stability explains more than its size. The gap between Omega-N and the battery stays between +0.073 and +0.144 while the battery itself moves from 0.09 to 0.41 across constructions that share no experimental technique. A margin that survives that much variation in the substrate is measuring something about the descriptor rather than about any one dataset.

5.2 Interpretation of the ties against recursive feature engines

Four ties against a recursive feature engine is the outcome the corollary of [1] predicted for the node-wise object, and it should be read as such rather than as a failure to win. The engine reaches the same accuracy using between 28 and 252 features generated by recursion; the descriptor reaches it with ten that have names. Where the two arms are combined the union wins, so they are complementary rather than substitutable.

The single win, on tolokors, is consistent with the same reading: tolokors has by far the highest mean degree in the set at 88.3, the richest triadic substrate, and the weakest attributes. It is the network where the conditions are most favourable, and the margin there is still only about two standard errors.

5.3 The two out-of-domain results

Both losses fall on the wrong side of a condition that is measurable before the comparison, and neither is anomalous.

On brazil-airports the degree carries 65% of the label entropy and classifies at 0.748 against a 0.267 majority baseline. A descriptor built to decorrelate from degree is answering a question the dataset does not pose. On roman-empire the mean degree is 2.9, close to a chain, and there is no triadic structure to localise.

The same condition appears a third time, in a different field and with a different mechanism, in the financial closure of Section 4.3: where the target is concentration itself, strength is a sufficient statistic and calibrating against a degree-preserving null removes precisely the signal being sought. Three independent appearances of one principle carry more weight than the same principle asserted once.

5.4 Conditions for use

The descriptor is worth reaching for when the graph is dense enough to have triadic structure, the target is not a function of connectivity, the graph semantics are co-occurrence rather than reachability, and node attributes are weak or absent. Where those hold it delivers ten interpretable coordinates without training. Where they do not, the right answer is often the degree, and where a learned representation is available and performance is the only criterion, the right answer is the embedding.

The same confound recurs wherever a construction parameter fixes density and local organisation together. In weighted gene co-expression analysis the soft-threshold power is chosen to force approximate scale-free topology, so the degree distribution and the local structure are set jointly before any biological question is asked, and the hub genes that follow are selected by connectivity; a recent assessment lists the scale-free assumption and the parameter tuning among that method's principal limitations, and reports datasets for which no parameter yields an adequate fit [40]. We have not tested the descriptor there and make no claim about it, but the shape of the problem is the one described above.

That boundary is not a hedge. On BioPlex the entire centrality battery sits at chance and Omega-N reaches 0.56, which is better and still not a tool. Most of the absolute performance on curated STRING is inherited from accumulated knowledge rather than discovered from structure. The descriptor reads graphs better; on a clean graph there is less to read.

5.5 Spectral-gap-matched null models for whole-brain modelling

The limitations above share a shape: the descriptor loses wherever the task is prediction and a richer representation exists. That suggests looking for tasks where the requirement is not prediction but the separation of two quantities that are ordinarily confounded, and where the argument must be made formally rather than empirically. Whole-brain modelling supplies one, and asks for it explicitly.

Deco and colleagues describe brain dynamics with models in which the operative structural quantity is the spectral gap of the coupling graph, constructing contrasting regimes by pruning k -regular graphs. Their own limitations state the difficulty: a denser or stiffer graph can improve functional-connectivity agreement for reasons unrelated to the mechanism, their two regimes differ in local edge density as a direct consequence of the pruning that defines them, and a null model matched for the spectral gap is required before the effect can be attributed to organisation rather than to the size of the gap [12].

That null is constructible, for the reason that motivates this paper. The spectral gap is a functional of the spectrum; per-node triangle counts are not [1]. Two graphs can therefore agree on the gap and disagree on local triadic organisation.

We verified this on their construction: 600 pruned k -regular graphs with $n = 40$ and $k = 20$. The fraction of variation not explained by the spectral gap is 62% for the per-node triadic excess and 19% for the clustering coefficient. The clustering coefficient is largely pinned by the gap, so matching on the gap nearly matches on clustering as well and clustering cannot serve as the contrast; the triadic excess can. Searching within a narrow gap window returns families differing by less than 0.05 in gap while differing two to three times as much in triadic excess. One matched pair has gaps of 2.62 and 2.45 and median triadic excesses of -0.48 and -0.06, a separation of roughly five standard deviations of the full distribution.

We report this as a construction, not as a result about the brain. Whether model fit changes when the gap is held fixed must be measured inside the whole-brain model, with the empirical data and the interference measure, and that is not done here. Both outcomes would be informative: if fit changes at fixed gap, the gap was not what drove it; if it does not change, the original interpretation survives an objection its authors anticipate. We note it because it is the one setting we have found in which the descriptor’s value lies in what can be proved about it rather than in what it predicts, and where a learned representation could not take its place: one cannot demonstrate what an embedding does and does not determine.

5.6 Performance attribution in graph rewiring

The same construction addresses an open question in a second and larger field. Graph rewiring for over-squashing proceeds by two families of methods: spectral approaches that add edges to increase the spectral gap, and curvature approaches that act on negatively curved edges, which in practice means acting on triangles. The two are not independent. A decrease in the number of triangles is accompanied by an increase in the spectral gap [39], so any rewiring step moves both quantities at once.

The consequence is an attribution problem that the field states in its own terms. A 2026 survey of the area reports that the benefits of curvature-based rewiring on real benchmarks are highly sensitive to training and rewiring hyperparameters and vary across graphs, and that reported gains often arise from favourable hyperparameter configurations rather than from consistent improvement over the original topology; it lists performance attribution among the open practical trade-offs [37]. A comparative study of rewiring methods through graph metrics reaches the same boundary from the other side, noting that an analysis of metrics can suggest correlation but not causation [38].

Families of graphs matched on the spectral gap and separated in triadic excess turn that correlation into a controlled comparison. Rewiring a benchmark graph to a target gap by two routes that differ in local triadic organisation, and measuring downstream accuracy on both, separates the contribution of the gap from the contribution of local structure. Neither outcome would be uninformative: if accuracy tracks the gap at fixed triadic organisation, spectral methods are measuring what they claim; if it tracks triadic organisation at fixed gap, the spectral account is incomplete.

We have not run that experiment and do not report it. We note it because the construction it requires is the one described above, and because the reason it is possible is the same property that motivates this paper: the spectral gap is a functional of the spectrum, and the per-node triangle count is not.

6. Limitations

The applicability rule was formulated after observing the two losses, and two statistics with two thresholds fitted on eight networks leave room to fit; what defends it is that both statistics are computed without reference to the descriptor’s performance, so a reader can apply them prospectively and be wrong, and we have not tested that prediction out of sample. The biological label is a proxy: approved-target status is not druggability, and both label sources carry study bias which we control three ways without removing. Excluding HuRI for lack of triadic substrate is defensible on a measurable criterion fixed in advance, but “the replication set does not suit my method” is what one says when a replication fails, and the exclusion should be weighed accordingly. The choice of a degree-preserving null is conventional rather than derived: such nulls remove mesoscopic structure, different randomisation schemes can reverse conclusions on the same data [17,18], and configuration models are generally adopted for tractability rather than on statistical evidence [19], so a richer null would absorb some of what we report as excess. Numerically, the spectral coordinate comes from an iterative solver that does not always converge and degenerates on disconnected graphs, and the curvature comparison subsamples on the four largest graphs. Everything here is undirected, unweighted and static.

Code and data availability

The reference implementation, the notebooks and the scripts that prepare each dataset are available at github.com/BiomeMakers/OmegaN. Datasets are obtained from their original sources and none are redistributed. The repository includes the pre-flight screening function that reports the two applicability statistics of Section 4.2, and the notebook that builds spectral-gap-matched graph families for the construction described in Section 5.5.

Declaration of LLM usage

The analyses, code and text of this manuscript were developed with the assistance of a large language model used interactively, for writing and debugging analysis scripts, searching and summarising literature, drafting and revising prose, and arguing against proposed interpretations. Every quantitative result was produced by code that was executed, and several conclusions in earlier drafts were reversed by that execution: a comparison withdrawn as unfair to a baseline turned out to have been generous to it, a result attributed to a novel graph construction turned out to depend on the length of a control window, and one coordinate of the descriptor was found to be inert on disconnected graphs. Responsibility for the claims, the design choices and any remaining errors rests with the author.

References

- [1] A. Acedo. The Functional Symbiotic Resilience Index: Topological Entropy, Wasserstein Curvature Bounds, and Non-Equilibrium Thermodynamics of Complex Networks. Preprint, 2026. github.com/BiomeMakers/OmegaS-fsri
- [2] L. Peel, J.-C. Delvenne, R. Lambiotte. Multiscale mixing patterns in networks. *PNAS* 115(16):4057-4062, 2018.
- [3] R. Guimera, L. A. N. Amaral. Functional cartography of complex metabolic networks. *Nature* 433:895-900, 2005.
- [4] K. Henderson et al. RolX: structural role extraction and mining in large graphs. *KDD*, 2012.
- [5] L. F. R. Ribeiro, P. H. P. Saverese, D. R. Figueiredo. struc2vec: learning node representations from structural identity. *KDD*, 2017.
- [6] C. Donnat, M. Zitnik, D. Hallac, J. Leskovec. Learning structural node embeddings via diffusion wavelets. *KDD*, 2018.
- [7] J. Jin, M. Heimann, D. Koutra. Toward understanding and evaluating structural node embeddings. *ACM TKDD*, 2021.
- [8] M. Piraveenan, M. Prokopenko, A. Y. Zomaya. Local assortativeness in scale-free networks. *EPL* 84(2):28002, 2008.

- [9] H. Wolf, L. Oeljeklaus, P. Kuehner, M. Grohe. Structural node embeddings with homomorphism counts. *arXiv:2308.15283*, 2023.
- [10] R. von Moos, M. Alain, B. Rieck. Invariant-based diagnostics for graph benchmarks. *arXiv:2605.06462*, 2026.
- [11] X. Ma, G. Qin, Z. Qiu, M. Zheng, Z. Wang. RiWalk: fast structural node embedding via role identification. *arXiv:1910.06541*.
- [12] G. Deco, Y. Sanz Perl, N. Greenstein, S. Chandaria, G. Scholes, M. L. Kringelbach. Quantum-like dynamics in whole-brain models of the human connectome. *Advanced Science*, e77103, 2026.
- [13] A. Acedo. Omega-S: a structural regulariser for large language model fine-tuning. *arXiv:2608.03887*.
- [14] O. Platonov et al. A critical look at the evaluation of GNNs under heterophily: are we really making progress? *ICLR*, 2023.
- [15] C. Ceylan, K. Ghooarchian, D. Kragic. Digraphwave: scalable extraction of structural node embeddings via diffusion on directed graphs. *arXiv:2207.10149*, 2022.
- [16] H. Li, S. Jiang, L. Zhang, S. Du, G. Ye, H. Chai. RAGFormer: learning semantic attributes and topological structure for fraud detection. *arXiv:2402.17472*, 2024.
- [17] J. Reichardt, R. Alamino, D. Saad. The interplay between microscopic and mesoscopic structures in complex networks. *PLoS ONE* 6(8):e21282, 2011.
- [18] B. Hao, I. A. Kovacs. Proper network randomization is key to assessing social balance. *Science Advances* 10(18):eadj0104, 2024.
- [19] L. Hebert-Dufresne, J.-G. Young, A. Daniels, A. Kirkley, A. Allard. Network compression with configuration models and the minimum description length. *Phys. Rev. E* 110:034305, 2024.
- [20] D. Ochoa et al. The next-generation Open Targets Platform: reimaged, redesigned, rebuilt. *Nucleic Acids Research* 51(D1):D1353-D1359, 2023.
- [21] M. Molloy, B. Reed. A critical point for random graphs with a given degree sequence. *Random Structures & Algorithms* 6(2-3):161-180, 1995.
- [22] M. Fiedler. Algebraic connectivity of graphs. *Czechoslovak Mathematical Journal* 23(2):298-305, 1973.
- [23] Y. Ollivier. Ricci curvature of Markov chains on metric spaces. *Journal of Functional Analysis* 256(3):810-864, 2009.
- [24] A. Grover, J. Leskovec. node2vec: scalable feature learning for networks. *KDD*, 2016.
- [25] R. N. Mantegna. Hierarchical structure in financial markets. *European Physical Journal B* 11:193-197, 1999.
- [26] M. Tumminello, T. Aste, T. Di Matteo, R. N. Mantegna. A tool for filtering information in complex systems. *PNAS* 102(30):10421-10426, 2005.
- [27] R. Milo, S. Shen-Orr, S. Itzkovitz, N. Kashtan, D. Chklovskii, U. Alon. Network motifs: simple building blocks of complex networks. *Science* 298(5594):824-827, 2002.
- [28] J. Tang, F. Hua, Z. Gao, P. Zhao, J. Li. GADBench: revisiting and benchmarking supervised graph anomaly detection. *NeurIPS Datasets and Benchmarks*, 2023.
- [29] L. Breiman. Random forests. *Machine Learning* 45(1):5-32, 2001.
- [30] T. Saito, M. Rehmsmeier. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE* 10(3):e0118432, 2015.
- [31] D. Szklarczyk et al. The STRING database in 2023: protein-protein association networks and functional enrichment analyses. *Nucleic Acids Research* 51(D1):D638-D646, 2023.
- [32] E. L. Huttlin et al. Dual proteome-scale networks reveal cell-specific remodeling of the human interactome. *Cell* 184(11):3022-3040, 2021.
- [33] K. Luck et al. A reference map of the human binary protein interactome. *Nature* 580:402-408, 2020.

- [34] Z. Zhu et al. Topological analysis of drug targets in the human protein-protein interaction network. *BMC Systems Biology*, 2012.
- [35] E. Guney, J. Menche, M. Vidal, A.-L. Barabasi. Network-based in silico drug efficacy screening. *Nature Communications* 7:10331, 2016.
- [36] S. L. Freshour et al. Integration of the Drug-Gene Interaction Database (DGIdb 4.0) with open crowdsource efforts. *Nucleic Acids Research* 49(D1):D1144-D1151, 2021.
- [37] H. Attali, N. Pernelle, D. Buscaldi, F. D. Malliaros. Graph rewiring in GNNs to mitigate over-squashing and over-smoothing: a survey. *IJCAI*, survey track, 2026. arXiv:2605.00951.
- [38] A. Benoit, C. Aitken, Y. He. Structural invariance matters: rethinking graph rewiring through graph metrics. arXiv:2510.20556, 2025.
- [39] S. Akansha. Over-squashing in graph neural networks: a comprehensive survey. *Neurocomputing* 642:130389, 2025.
- [40] T. Osoné, T. Takao, S. Otake, T. Takarada. SGCRNA: spectral clustering-guided co-expression network analysis without scale-free constraints for multi-omic data. *Briefings in Bioinformatics* 27(1):bbag021, 2026.
- [41] B. Kaminski, L. Kraininski, P. Pralat, F. Theberge. Unsupervised framework for evaluating and explaining structural node embeddings of graphs. arXiv:2306.10770, 2023.